

SUMMARY

U.S. Citizen and Senior Software Engineer specializing in data-driven systems and real-time decision platforms for large-scale, mission-critical operations. Proven experience designing and operating real-time analytics pipelines, telemetry systems, and automated decision loops that translate data signals into production actions. Strong background in backend and distributed systems using Java and Python, with hands-on experience building scalable systems that integrate data, models, and operational workflows. Experienced in defining decision metrics, operational signals, and building data-driven control loops to support large-scale optimization, automation, and decision-making in complex environments.

WORK EXPERIENCE

Walmart, Senior Data Analyst / Software Engineer – Last-Mile Delivery **Oct. 2023 – Apr. 2026**

- Contributed core real-time decision signals, analytics dashboards (Tableau), and automation pipelines to a large-scale, data-driven decision system supporting Walmart last-mile delivery operations, enabling dynamic capacity management, incentive deployment, and zone-level investment decisions.
- Defined and operationalized a core real-time decision metric (“Demand–Supply Gap”) to quantify imbalance between trip demand, order volume, and active driver supply at the delivery-zone level, establishing it as a primary signal used by operations, analytics, and data science teams.
- Integrated Demand–Supply Gap with near real-time weather risk signals to identify environment-driven operational risk, enabling proactive mitigation actions such as capacity throttling, targeted incentive deployment, and special zone monitoring.
- Connected these decision signals to downstream data science models and decision frameworks that dynamically adjusted delivery capacity, deployed incentives, and prioritized zone-level operational investments.
- Built and operated the supporting backend platform, maintaining and evolving ~20 microservices and distributed data systems; developed backend APIs and fallback tooling to support driver assignment, trip validation, and outage recovery workflows.
- Implemented real-time monitoring, diagnostics, and telemetry pipelines, including self-service diagnostic APIs and data pipelines, reducing data latency by ~60% and improving SLA visibility across critical services.
- Improved system reliability by tuning alerting thresholds and signal quality across multiple services, reducing false positives by ~30% and enabling faster incident detection and recovery.
- Delivered end-to-end automation pipelines that monitored upstream data updates, identified actionable trips via targeted queries, and safely invoked backend APIs to apply automated mitigation actions in production.

KLA-Tencor, Application Development Engineer **Oct. 2021 - Jun. 2022**

- Developed Python-based automation tools on Unix systems to streamline engineering workflows, reducing analysis time by ~30% and improving consistency across test environments. Collaborated with cross-functional engineering teams to productionize internal tools and incorporate user feedback into iterative improvements.

Shanghai Genius Academy, Co-Founder/Founding Software Engineer **Aug. 2020 - Oct. 2023**

- Built and operated internal business systems using React, FastAPI, and AWS, including payroll automation and operational portals, reducing manual processing overhead by ~60%.
- Designed API-first services and modular workflows with production monitoring and automated testing, supporting reliable day-to-day operations and system maintainability.

Purdue University, Graduate Research Assistant **Aug. 2018 - May 2020**

- Conducted graduate research on magnetic thin-film materials, including nanoscale fabrication, thin-film deposition, and experimental characterization for magnetophotonic and spintronic applications.

SKILLS

- Programming Languages: Java, Python
- Data & Decision Systems: Real-time analytics, decision signals, anomaly detection, decision support systems
- Backend & Distributed Systems: Microservices, REST APIs, distributed systems, fault-tolerant design
- Data Platform & Analytics: BigQuery, Tableau, Airflow
- Observability & Reliability: Telemetry, logging, metrics, alerting, incident response

EDUCATION

The University of Texas at Austin | *M.S. in Artificial Intelligence (in progress)*

Aug. 2024 - Present

Purdue University | *M.S. in Materials Engineering*

Graduated 2020

National Tsing Hua University | *B.S./M.S. in Chemical Engineering*

Graduated 2016